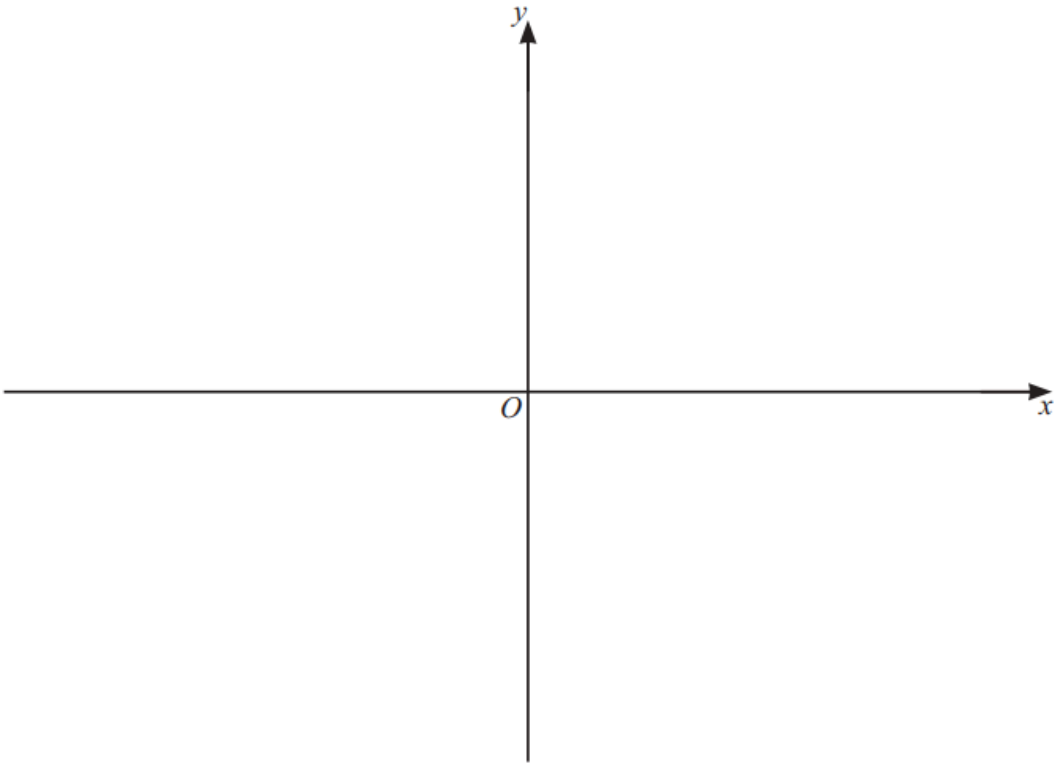
Additional Mathematics (0606)

Question Paper 1 (Practice) **Non-Calculator**
Chapter 1: Functions **Sources:** CIE 0606 Past Papers
Compiler: Julian **Quality Control**: Julian

Question 1:

On the axes, sketch the graphs of $y = 4|x-1|$ and $y = |3x+2|$, stating the intercepts with the axes. [3]



Question 2:

(a) Write $2x^2 - 2x + 3$ in the form $a(x+b)^2 + c$, where a, b and c are constants. [3]

It is given that $f(x) = 2x^2 - 2x + 3$, for $x \leqslant p$.

(b) Write down the greatest value of p for which f has an inverse. [1]

(c) Using this value of p, write down the range of f. [1]

(d) Using this value of p, find an expression for f^{-1}. [3]

Question 3:

Functions f and g are such that

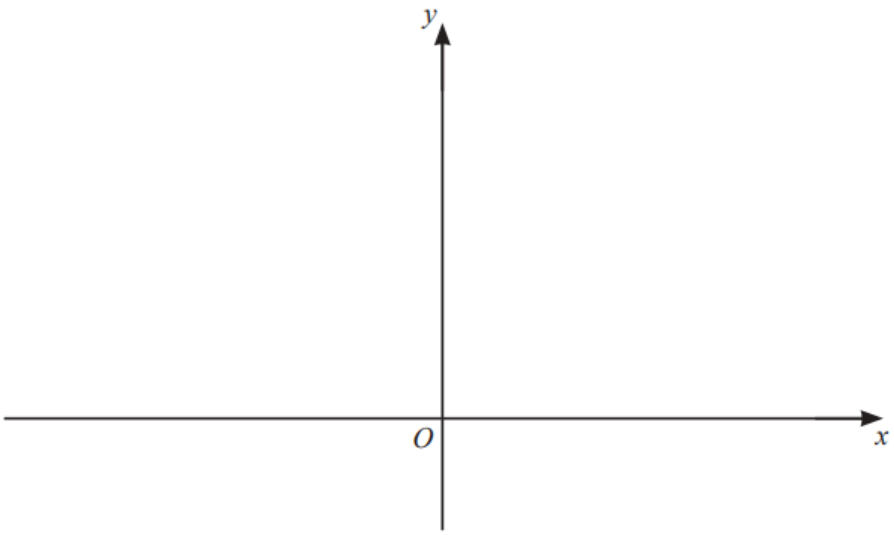
$$f(x) = \frac{3x}{x+4} \quad \text{for } x > 0$$

$$g(x) = \sqrt{x+2} \quad \text{for } x > -2.$$

Solve the equation $fg(x) = 1$. [4]

Question 4:

(a) On the axes, sketch the graph of $y = |4x - 6|$, showing the points where the graph meets the axes. [2]



Question 5:

The functions f and g are defined by

$$\mathrm{f}(x) = \frac{3x^2}{4x-1} \quad \text{for } x < 0$$

$$\mathrm{g}(x) = \frac{1}{x^2} \quad \text{for } x < 0.$$

(a) Explain why the function fg does **not** exist. [1]

(b) Given that the function gf does exist, find and simplify an expression for $\mathrm{gf}(x)$. [2]

(c) Show that $\mathrm{f}^{-1}(x)$ can be written as $\dfrac{px - \sqrt{x(qx+r)}}{3}$ where p, q and r are integers. [4]

Question 6:

(b) The function g is such that $g : x \mapsto x^{\frac{1}{2}} - 4$, for $x > 0$. Solve the equation $g^2(x) = -2$. [3]

Question 7:

Solve the equation $4|7x - 3| - 5 = 9$. [3]

Question 8:

(a) It is given that $\mathrm{f} : x \to 2x^2$ for $x \geqslant 0$,

$\mathrm{g} : x \to 2x+1$ for $x \geqslant 0$.

Each of the expressions in the table can be written as one of the following.

f' f'' g' g'' fg gf f^2 g^2 f^{-1} g^{-1}

Complete the table. The first row has been completed for you. [5]

Expression	Function notation
2	g'
0	
$4x$	
$8x^2+8x+2$	
$4x+3$	
$\frac{x-1}{2}$	

(b) It is given that $\mathrm{h}(x) = (x-1)^2+3$ for $x \geqslant a$. The value of a is as small as possible such that h^{-1} exists.

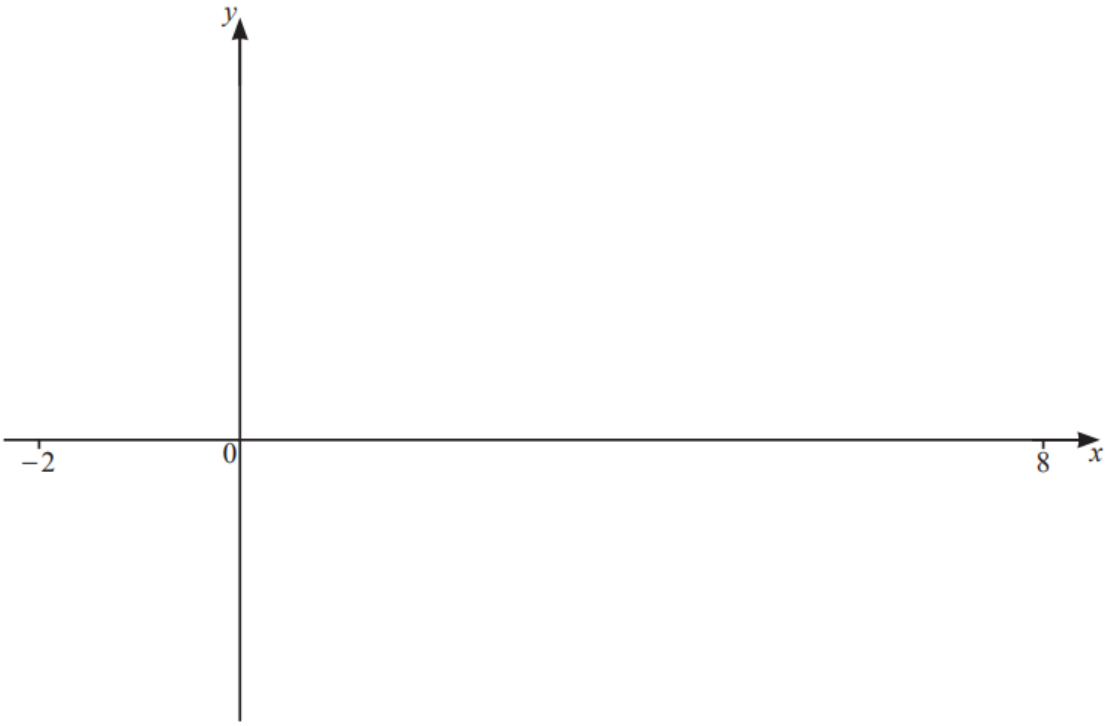
(i) Write down the value of a. [1]

(ii) Write down the range of h. [1]

(iii) Find $h^{-1}(x)$ and state its domain. [3]

Question 9:

(a) On the axes, sketch the graphs of $y = |2x+1|$ and $y = |5-3x|$ for $-2 \leqslant x \leqslant 8$. State the coordinates of the points where these graphs meet the coordinate axes. [3]



Question 10:

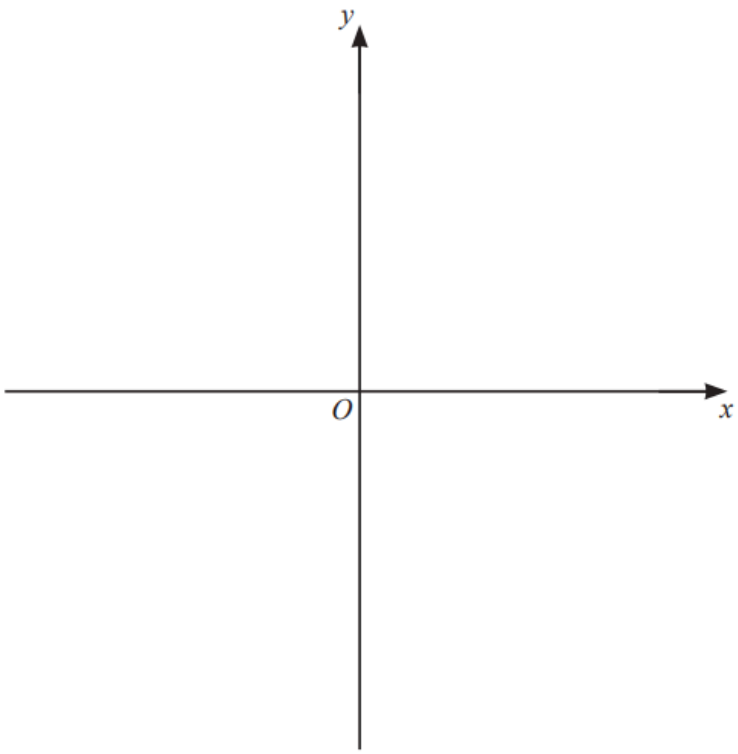
The functions f and g are defined as follows.

$$f(x) = x^2 + 4x \quad \text{for} \quad x \in \mathbb{R}$$

(a) Find the range of f. [2]

Question 11:

(a) On the axes below, sketch the graph of $y = |5x - 7|$, showing the coordinates of the points where the graph meets the coordinate axes. [3]



(b) Solve $5|5x - 7| - 1 = 14$. [3]

Question 12:

$$\mathrm{f} : x \mapsto (2x+3)^2 \quad \text{for } x > 0$$

(a) Find the range of f. [1]

(b) Explain why f has an inverse. [1]

(c) Find f^{-1}. [3]

(d) State the domain of f^{-1}. [1]

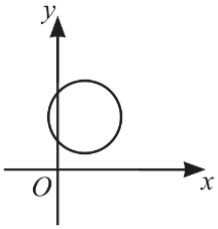
Question 13:

The function g is defined by $g(x) = (x-4)^2 + 1$ for $x \leqslant 4$.

(b) **(i)** Find an expression for $g^{-1}(x)$ and state its domain and range. [4]

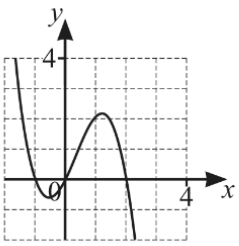
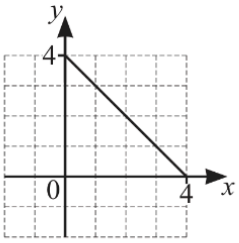
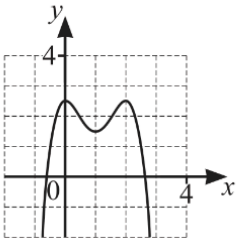
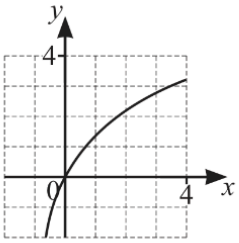
Question 14:

(a)



Explain why this graph does **not** represent a function. [1]

(b) The table shows the graphs of four different functions.

	is one-one	is many-one	is its own inverse
			
			
			
			

Tick (✓) each correct box in the table.
There may be more than one tick in a row or a column. [4]

(c) Functions f and g are defined as follows.

$\mathrm{f}: x \mapsto \sin x \qquad \text{for } 30° \leqslant x \leqslant a°$

$\mathrm{g}: x \mapsto \sqrt{x - \frac{1}{2}} \qquad \text{for } x \geqslant \frac{1}{2}$

It is given that the function gf exists.

(i) Find the value of a so that the domain of gf is as large as possible.

You may use the information that $\sin 30° = \frac{1}{2}$. [2]

(ii) For the domain found in **part (i)**, find the range of the function gf. [2]

(iii) Determine whether the function g^2 exists. [1]

Question 15:

$$f(x) = x^2 + 2x - 3 \quad \text{for } x \geqslant -1$$

(a) Given that the minimum value of $x^2 + 2x - 3$ occurs when $x = -1$, explain why $f(x)$ has an inverse. [1]

Question 16:

(b)

$$g(x) = x + 5 \quad \text{for } x \in \mathbb{R}$$

$$h(x) = \sqrt{2x - 3} \quad \text{for } x \geqslant \frac{3}{2}$$

Solve $gh(x) = 7$. [3]

Question 17:

Solve $|3x-2| = 4+x$. [3]

Question 18:

(a) It is given that $g(x) = 6x^4 + 5$ for all real x.

(i) Explain why g is a function but does not have an inverse. [2]

(ii) Find $g^2(x)$ and state its domain. [2]

It is given that $h(x) = 6x^4 + 5$ for $x \leqslant k$.

(iii) State the greatest value of k such that h^{-1} exists. [1]

(iv) For this value of k, find $h^{-1}(x)$. [3]

Question 19:

It is given also that $g(x) = x^2 + 4$ for $x \in \mathbb{R}$.

Find the exact solutions of $g^2(x) = 40$. [3]

Question 20:

(a) It is given that

$$f : x \mapsto \sqrt{x} \quad \text{for} \quad x \geqslant 0,$$
$$g : x \mapsto x + 5 \quad \text{for} \quad x \geqslant 0.$$

Identify each of the following functions with one of f^{-1}, g^{-1}, fg, gf, f^2, g^2.

(i) $\sqrt{x+5}$ [1]

(ii) $x - 5$ [1]

(iii) x^2 [1]

(iv) $x+10$ [1]

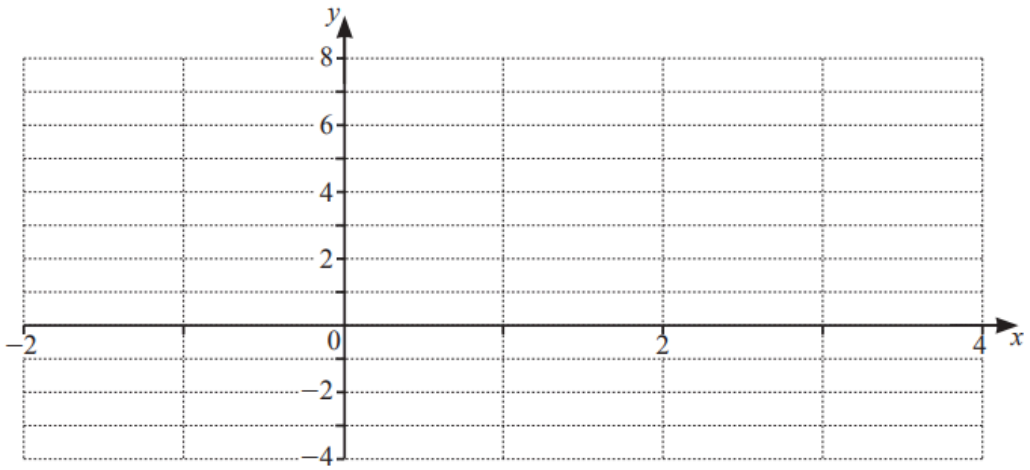
(b) It is given that $h(x) = a + \frac{b}{x^2}$ where a and b are constants.

(i) Why is $-2 \leqslant x \leqslant 2$ not a suitable domain for $h(x)$? [1]

(ii) Given that $h(1) = 4$ and $h'(1) = 16$, find the value of a and of b. [2]

Question 21:

(i) On the axes below, draw the graph of $y = |2x-3|$.



[2]

(ii) Solve the equation $7-|2x-3|=0$. [3]

Question 22:

The function f is defined by $f(x)=\dfrac{1}{2x-5}$ for $x>2.5$.

(i) Find an expression for $f^{-1}(x)$. [2]

(ii) State the domain of $f^{-1}(x)$. [1]

(iii) Find an expression for $f^{2}(x)$, giving your answer in the form $\dfrac{ax+b}{cx+d}$, where a, b, c and d are integers to be found. [3]

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Total Marks: 107

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